

Factors Associated with Laboratory Confirmation of Ebola Virus Disease Among Suspected Cases in Sierra-Leone, West Africa:

A Retrospective Secondary Data Analysis

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Abstract

Ebola virus disease(EVD) remains a major public health threat in Africa, with recurrent outbreak response. Understanding factors associated with laboratory-confirmed Ebola cases among suspected cases is useful for improving case detection and control measures. This study assessed the factors associated with laboratory-confirmed Ebola virus disease among suspected cases in Sierra-Leone.

A retrospective cross-sectional study was conducted using secondary Ebola surveillance data collected during West Africa Ebola Outbreak. Data on age, sex, district of residence, and Ebola confirmation status were analyzed using R statistical software. Descriptive statistics were used to summarize participant characteristics. Pearson's chi-square test assessed associations between the independent variables and Ebola confirmation status. Binary logistic regression was performed to identify the independent predictors of laboratory-confirmed Ebola virus disease. Crude and adjusted odds ratios (ORs) with 95% confidence intervals (CIs) were reported.

A total of 196 suspected Ebola cases were included in the analysis, of which 179(91.3%) were laboratory confirmed. Chi-square analysis showed a significant association between district of residence and Ebola confirmation status ($\chi^2 = 11.33, p = 0.003$), while sex and age group were not significantly associated. In multivariate logistic regression analysis, suspected cases from district categorized as "Other District" had significantly lower odds of laboratory-confirmed Ebola compared to those from Kailahun District ($AOR = 0.11, 95\%CI: 0.02 - 0.56, p = 0.007$). Sex and age group were not independently associated with Ebola confirmation status.

District of residence was the only independent predictor of laboratory-confirmed Ebola virus disease among suspected cases. These findings highlight the importance of geographical factors in Ebola transmission and emphasize the need for strengthened surveillance, targeted interventions, and outbreak preparedness in high-risk districts.

Keywords: Ebola Virus Disease, Surveillance, Logistic Regression, Outbreak District, West Africa

Dedication

I dedicate this paper work to my family for their unwavering support, encouragement and sacrifice throughout my academic journey. Their belief in my ability has been a constant source of motivation and inspiration.

Additionally I dedicate this work to all healthcare workers, and public health professionals whose commitment to disease surveillance and outbreak response continues to contribute towards the prevention and control of infectious disease globally.

Acknowledgments

I would like to express my sincere gratitude to all individuals and institutions whose publicly available, resources, training materials and scientific publication contributed to the successful completion of this study. I am also grateful to the public health professionals and surveillance teams which were involved in the collection and management of Ebola surveillance data during the outbreak period.

This study was conducted independently as part of my academic and professional development in epidemiology and public health.

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Introduction

0.1 Introduction

0.1.1 Ebola Background

Ebola Virus Disease is a severe and often deadly zoonotic disease caused by viruses belonging to the genus Ebola virus of the Filoviridae family Malvy et al. (2019); Jacob et al. (2020). The disease is characterized by sudden onset of fever, muscle pain, fatigue, vomiting, diarrhoea and in severe cases, hemorrhagic manifestations and failure of many organs Nsio et al. (2023); ?. Human to Human transmission occur through direct contact with blood, body-fluid, secretions and contaminated materials from the infected victims and through contact with infected animals Rewar and Mirdha (2014); Judson et al. (2015). Ebola Virus Disease has been associated with high case fatality rates in the range of 25% to 90%, following on the outbreak context, healthcare access, and timely intervention Chertow (2019). The 2014 – 2016 West African Ebola outbreak which affected Guinea, Liberia, and Sierra-Leone was among the largest and most complex Ebola epidemics ever recorded, resulting in 28,000 cases reported and over 11000 deaths reported according to World Health Organization (WHO) Kamorudeen et al. (2020); Kamradt-Scott (2016). The outbreak overwhelmed the fragile Health Care systems, interfered with the daily healthy service, and highlighted the major gap in disease surveillance, early diagnosis, and outbreak preparedness. Understanding factors associated with Laboratory-confirmed Ebola cases among suspected individuals remains crucial for tightening the surveillance systems, improving case identification, and guiding targeted public health interventions during future outbreaks.

0.1.2 Epidemiological Importance

Controlling of Ebola Virus Disease outbreaks remains a major public challenge, especially in the resource-limited settings, where health care systems may be fragile and surveillance infrastructure inadequate. Late case detection, limited laboratory capacity, poor contact tracing, and community mistrust can lead to rapid disease transmission and increased mortality rate during outbreaks. Early Identification and isolation of infected victims are therefore crucial components of outbreak containment strategies. Surveillance system plays a critical role in detecting suspected cases, monitoring transmission trends, guiding public health response, and supporting timely laboratory confirmation of Ebola Infection Tambo et al. (2014); Bell (2016). However, not all the suspected cases are ultimately confirmed as Ebola Virus disease victims, as many individuals may show up the symptoms similar to other infectious diseases such as malaria, typhoid fever or lassa fever. Differentiating confirmed from sus-

pected cases is epidemiologically important because it helps more accurately allocate healthcare resources, improve outbreak monitoring and enhances the effectiveness of response interventions. Knowing demographic and geographical factors associated with laboratory confirmation among suspected Ebola cases may therefore be useful in strengthening surveillance efficiency and improving outbreak preparedness in high risk settings. Alonge et al. (2019).

0.1.3 Knowledge Gap

Although much effort was made during West Africa Ebola outbreak, limited evidence exists regarding the factors associated with laboratory confirmation among suspected cases in Sierra-Leone.

0.1.4 Study Objectives

General Objective

To determine the factors associated with laboratory-confirmed Ebola Virus disease among suspected cases in Sierra-Leone.

Specific Objectives

1. To describe the demographic and epidemiological characteristics of suspected Ebola virus disease cases in Sierra-Leone.
2. To assess the relationship between demographic variables and Ebola confirmation status among suspected cases.
3. To determine the independent predictors of laboratory-confirmed Ebola virus disease among suspected cases using binary logistic regression analysis.

Methodology

0.2 Methodology

0.2.1 Study Design

A retrospective cross-sectional study was carried out using secondary Ebola surveillance data to assess factors related to laboratory-confirmed Ebola virus disease among suspected cases in Sierra-Leone.

0.2.2 Study Setting

The study was carried out in Sierra-Leone, one of the West African countries affected during the 2014 – 2016 outbreak of the Ebola virus. The incident affected many districts across the country, leading to the substantial morbidity, mortality, and disruption of healthcare services. Surveillance data from reported districts included in the national Ebola response system were analyzed. The study time covered suspected Ebola cases reported during the outbreak period from May-2014 to July-2014

0.2.3 Data Source

The study data were obtained from Ebola Surveillance records maintained by the ministry of health during the outbreak period. The surveillance database contained routinely collected information on suspected Ebola cases identified through the national disease surveillance and response system.

0.2.4 Study Variables

The primary outcome variables were the laboratory confirmation status of Ebola virus disease, categorized as:

- Confirmed case
- Suspected case

The Independent variables Included

- Age
- Sex
- District
- Delay between symptom onset and sample collection

Delay in sample collection was calculated as the number of days between the onset date of symptom and sample collection.

0.2.5 Data Cleaning and Management

Data cleaning and management were performed using R statistical software. Duplicated records were removed using separate identification numbers. Missing values and inconsistencies were assessed prior to analysis. Date variables were changed to standard date formats to ease the calculation of epidemiological indicators and reporting delays. Categorical variables including sex, district, and Ebola confirmation status were changed to factor variables for statistical analysis. Age variable were also checked for plausibility, and the records with invalid and inconsistent dates were excluded from the analysis.

0.2.6 Statistical Analysis

Descriptive Analysis

Descriptive statistics were used to summarize demographic and epidemiological characteristics of suspected Ebola cases. Frequencies and percentages were calculated for categorical variables, whereas median and inter-quartile ranges were used to summarize continuous variables such as age and reporting delay due to non-normal distribution.

Table 1 presents the descriptive characteristics of suspected Ebola cases included in the Analysis.

Figure 1 presents the epidemic curve showing the temporal distribution of Ebola cases during the outbreak period.

Bivariate Analysis

Pearson’s Chi-Square test was used to assess the relationship between categorical independent variables and Ebola confirmation status. Variables examined included sex, age group and district.

Table 2 presents the results of the Chi-Square analysis.

Logistic Regression Analysis

Binary logistic regression analysis was performed to examine factors independently associated with laboratory-confirmed Ebola Virus disease among the suspected cases. Variables with p-Values less than 0.20 at bivariate analysis were included in the multivariate logistic regression model.

The logistic regression model was defined as;

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K$$

where (p) represents the probability of laboratory-confirmed Ebola Virus disease.

Crude odds ratios (*COD*), adjusted odds ratios (*AOR*), 95% confidence interval (*CI*), and p-Values were reported . The statistical significance was set at $p < 0.05$

Table 3 presents the results of the binary logistic regression analysis.

0.2.7 Ethical Considerations

Ethical approval for the study was obtained . Secondary de-identified surveillance data were used for Analysis and no personally identifiable data was included in the data-set. Data confidentiality and privacy were maintained throughout the study.

Results

0.3 Results

A total number of 200 Ebola surveillance records were retrieved from the national surveillance database for the study period. After data cleaning and elimination of duplicates and incomplete data records, 196 suspected Ebola cases were retained for final analysis.

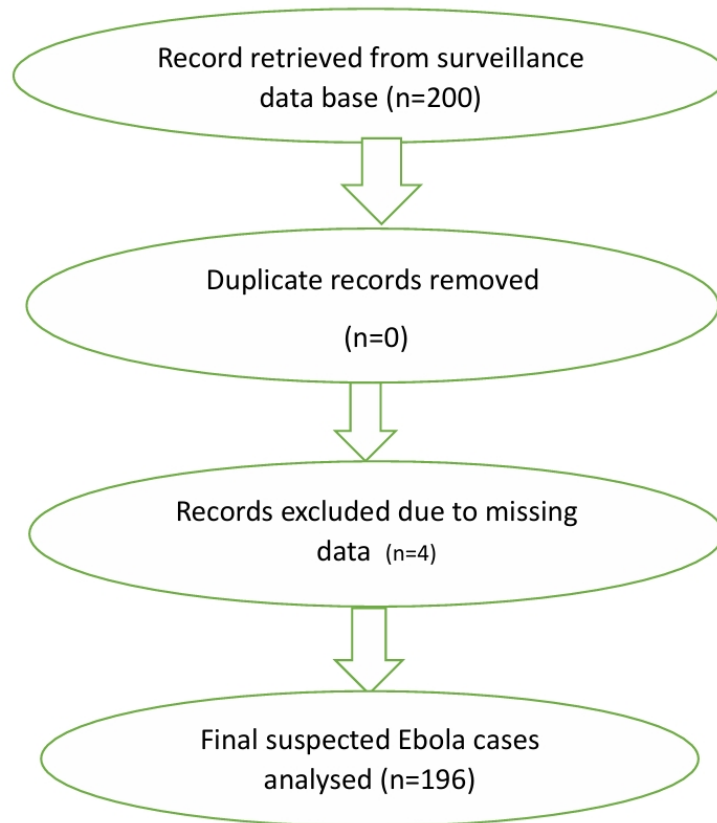


Figure 1: Flow Diagram of study participants

0.3.1 Descriptive Characteristics of Suspected Ebola Cases in Sierra-Leone

The total number of 196 suspected Ebola cases were analyzed. Males accounted for 42.857% of the total participants, while females represented 57.143% . The median age of participants was 35 years (IQR:25). The majority of suspected cases originated from Kailahun district with 152 cases, followed by Kenema district with 33 Cases.

Variables	Categories	Frequency(n)	Percentage(%)
Sex	Male	84	42.857
	Female	112	57.143
Age Group	0-14	31	15.82
	15-24	26	13.27
	25-34	37	18.88
	35-44	48	24.49
	45+	54	27.55
Ebola Status	Confirmed	179	91.33
	Suspected	17	8.67
Districts	Kailahun	152	77.5
	Kenema	33	16.84
	Western Urban	4	2.04
	Bo	2	1.02
	Kono	2	1.02
	Port Loko	2	1.02
	Kambia	1	0.51

Table 1: Descriptive Statistics

Temporal Distribution of Ebola Cases

The epidemic curve demonstrated changes in the number of suspected Ebola cases reported during the outbreak period.

Reported cases increased progressively during the outbreak and peaked during epidemiological week 24 , after which a gradual decline was observed. Confirmed Ebola Cases followed a similar temporal trend, suggesting active transmission during the peak outbreak period.

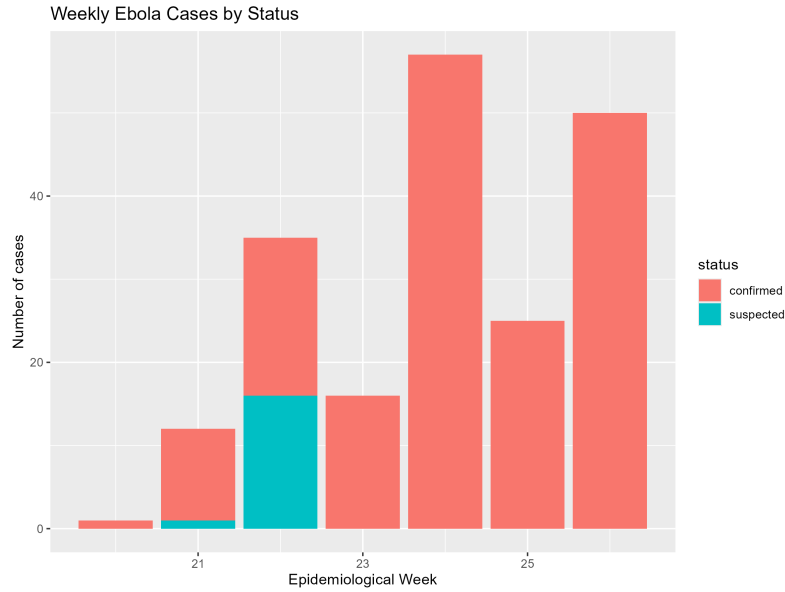


Figure 2: Weekly Distribution of Suspected and Confirmed Ebola Cases

0.3.2 Bivariate Analysis

Pearson's Chi-Square test was conducted to assess the relationship between selected demographic variables and Ebola confirmation status among suspected cases.

Variables	Categories	Confirmed n(%)	Suspected n(%)	χ^2	p-Value
Sex	Male	79 (94.05%)	5 (5.95%)	0.83867	0.3598
	Female	100 (89.29%)	12 (10.71%)		
Age Group	0-14	29 (93.55%)	2 (6.45%)	3.8901	0.4211
	15-24	24 (92.31%)	2 (7.69%)		
	25-34	32 (86.49%)	5 (13.51%)		
	35-44	42 (87.5%)	6 (12.5%)		
	45+	52 (96.29%)	2 (3.7%)		
Districts	Kailahun	141 (92.76%)	11 (7.24%)	11.328	0.00346
	Kenema	31 (93.94%)	2 (6.06%)		
	Others	7 (63.64%)	4 (36.36%)		

Table 2: Chi-Square Test

Pearson's Chi-Square test was performed to assess the relationship between demographic characteristics and Ebola confirmation status among suspected cases. No

statistically significant association was observed between sex and Ebola confirmation status ($\chi^2 = 0.84, p = 0.360$) or between age group and Ebola confirmation status ($\chi^2 = 3.89, p = 0.421$). However, district of residence was significantly associated with Ebola confirmation status ($\chi^2 = 11.33, p = 0.003$). A higher proportion of suspected cases from Kailahun (92.8%) and Kenema (93.9%) were laboratory confirmed compared to cases from other districts (63.6%).

0.3.3 Logistic Regression Analysis

Variable	Category	COR (95% CI)	p-Value	AOR (95% CI)	p-Value
Sex	Male Vs Female	1.9 (0.67 - 6.16)	0.247	2.78 (0.88 - 10.94)	0.102
Age Group	15-24 Vs 0-14	0.83 (0.09 - 7.32)	0.855	0.90 (0.10 - 8.15)	0.92
	25-34 Vs 0-14	0.44 (0.06 - 2.22)	0.350	0.51 (0.07 - 2.77)	0.45
	35-44 Vs 0-14	0.48 (0.07 - 2.27)	0.392	0.66 (0.09 - 3.29)	0.63
	45+ Vs 0-14	1.79 (0.21 - 15.57)	0.569	1.76 (0.20 - 15.54)	0.59
District	Kenema Vs Kailahun	1.21 (0.30 - 8.10)	0.81	1.29 (0.32 - 8.78)	0.75
	Other Vs Kailahun	0.14 (0.04 - 0.58)	0.0045	0.11 (0.02 - 0.56)	0.007

Table 3: Logistic Regression Analysis

Table 3 shows the crude and adjusted logistic regression analysis of factors associated with laboratory-confirmed Ebola virus disease among suspected cases.

After some adjustment for age group, sex, and district, sex was not significantly associated with Ebola confirmation status. Male suspected cases had higher odds of laboratory-confirmed Ebola compared to females ($AOR = 2.78$, 95% CI: 0.88–10.94), however this association was not statistically significant ($p = 0.102$).

Similarly, age group was not independently associated with Ebola confirmation status. Compared with individuals aged 0–14 years, those aged 15–24 years ($AOR = 0.90$, 95% CI: 0.10 – 8.15, $p = 0.92$), 25 – 34 years ($AOR = 0.51$, 95%, $p = 0.45$), and 35 – 44 years ($AOR = 0.66$, 95% CI: 0.09 – 3.29, $p = 0.63$) had lower odds of Ebola confirmation. Individuals aged 45 years and above had higher odds of Ebola confirmation ($AOR = 1.76$, 95% CI: 0.20 – 15.54, $p = 0.59$). However, non of these association reached statistical significance.

District of residence was independently associated with Ebola confirmation status. Compared with suspected cases from Kailahun District, cases from Kenema district had similar odds of Ebola confirmation ($AOR = 1.29$, 95% CI: 0.32 – 8.78, $p = 0.75$). In contrast, suspected cases from the grouped "Other District" category, had significantly lower odds of laboratory-confirmed Ebola compared to those from Kailahun District ($AOR = 0.11$, 95% CI: 0.02 – 0.56, $p = 0.007$). This indicates that suspected cases reported from districts outside the major outbreak districts

were approximately 89% less likely to be laboratory-confirmed for Ebola than those reported from Kailahun.

Overall, district of residence was the only factor that remained significantly associated with Ebola confirmation after controlling for potential confounding variables.

Discussion

0.4 Discussion

This study assessed factors associated with laboratory-confirmed Ebola virus disease among suspected cases in Sierra-Leone using the secondary Surveillance data collected during the west African Ebola outbreak. Of the 196 suspected Ebola cases included in the analysis, majority were laboratory confirmed, reflecting the high risks of Ebola transmission during outbreak period. The study found that district of residence was significantly associated with Ebola confirmation status, while sex, and age group were not independently associated with laboratory-confirmation of Ebola infection.

In the bivariate analysis, district of residence demonstrated a statistically significant association with Ebola confirmation status. This association remained significant in the multivariate logistic regression model after adjusting for sex and age group. Specifically, suspected cases from districts grouped as "Other District" had significantly lower odds of Ebola confirmation compared to those from Kailahun District. This result suggests that the geographical location played an important role in finding the likelihood of Ebola infection during the outbreak.

The observed geographical variation is consistent with reports from West Africa Ebola outbreak, which identified certain districts as major transmission hotspots. Kailahun District was among the earliest and most heavily affected districts during the outbreak and served as a crucial epicenter for Ebola transmission. Higher confirmation rates in Kailahun may therefore reflect bigger community transmission, increased exposure to the infected individuals, and sustained chains of transmission compared to districts that experienced fewer cases. Similar geographical clustering of Ebola cases has been reported in outbreak investigations and Surveillance studies conducted in Sierra-Leone, Guinea, and Liberia, where transmission varied considerably across districts Suchar et al. (2018).

Although male suspected cases had higher odds of laboratory-confirmed Ebola compared to females, the association was not statistically significant after adjusting the potential confounders. This finding differed from other outbreak studies that reported increased Ebola risks among males possibly due to occupational exposure, mobility patterns, and involvement in activities that increase contact with infected individuals. However, the absence of a significant association in the present study shows that both males and females were similarly affected during the outbreak. This may reflect widespread community transmission where exposure occurs across households and communities regardless of sex.

In the similar way age group was not significantly associated with Ebola confirmation status. Although individuals aged 45 years and above showed higher odds of confirmation compared to younger age group, the observed association was not statistically significant. Previous studies have reported varying relationships between

age and Ebola infection, with some indicating higher risks among adults due to care-giving responsibilities, Occupational exposure, and participation in the burial practices. The lack of statistical significance in the present study may indicate that the Ebola Exposure was widely spread across age group during the outbreak period.

0.4.1 Public Health Implication

The findings of this study have important implications for Ebola Surveillance and outbreak response. The strong association between district of residence and Ebola confirmation highlights the importance of strengthening district level surveillance systems, particularly in areas identified as transmission hotspots. Enhanced case detection, timely laboratory testing and rapid response mechanisms should be prioritized in districts with high risks of Ebola to facilitate early outbreak control.

Targeted public health interventions should also focus on geographical area with increased transmission risk. During future outbreaks, resources such as diagnostic capacity, contact tracing teams, infection prevention supplies, and community engagement activities should be strategically allocated to districts with elevated risk profiles.

Furthermore, the findings emphasize the importance of outbreak preparedness at the district level. Strengthening routine disease surveillance, improving laboratory networks, training healthcare workers, and promoting community awareness can enhance the capacity of health systems to monitor and respond to Ebola outbreaks promptly.

Overall, this study demonstrates that geographical location was the primary factor associated with laboratory-confirmed Ebola virus disease among suspected cases. Therefore, Strengthening surveillance and preparedness effort in high risk districts may contribute positively to reducing Ebola transmission and improving outbreak response outcomes in future Epidemics.

Strengths and Limitations

0.5 Strengths and Limitations

0.5.1 Strengths

This study used real world Ebola surveillance data collected during the actual outbreak, thereby enhancing the epidemiological relevance and public health significance of the findings. The use of routinely collected surveillance data provides meaningful insights into factors associated with laboratory-confirmed Ebola cases under real outbreak conditions.

Furthermore, the application of multivariate logistic regression enabled adjustment for potential confounding factors and allowed identification of independent predictors of Ebola confirmation.

0.5.2 Limitations

This study has numerous limitations. Firstly, the analysis was based on secondary surveillance data, which limited the variables available for investigation.

Important factors such as contact history, occupation, clinical symptoms, and treatment information were not included in the data-set and therefore could not be assessed. Secondly, the relatively small number of non-confirmed cases may have reduced the statistical power of the study and contributed to wide confidence intervals for some estimates. Lastly, surveillance data are subjected to the reporting and recording errors, which may have introduced information bias and affected the accuracy of some findings.

Conclusion

0.6 Conclusion

This study assessed factors associated with laboratory-confirmed Ebola virus disease among suspected cases in Sierra-Leone using secondary surveillance data. District of residence was the only factor independently associated with Ebola confirmation status. Suspected cases from district outside the major outbreak areas had significantly lower odds of laboratory-confirmed Ebola compared to those from Kailahun District. Sex and age group were not significantly associated with Ebola confirmation after adjustment for potential confounders.

These findings highlight the importance of geographical location in Ebola transmission dynamics and underscore the need for strengthened surveillance, rapid case detection, and targeted outbreak response strategies in high-risk districts. Enhancing district level preparedness and surveillance capacity may contribute to improving control of future Ebola outbreaks.

Recommendation

0.7 Recommendation

1. Strengthen Ebola surveillance and early case detection system in high-risk districts to facilitate rapid identification and response to outbreaks.
2. Prioritize allocation of outbreak response resources, including laboratory testing, contact tracing, and infection prevention measures, to districts with historically high Ebola transmission.
3. Enhance district-level outbreak preparedness through regular training of health-care workers and strengthening emergence response capacity.
4. Future studies should incorporate additional exposure and clinical variables to better understand factors associated with Ebola virus disease confirmation.

0.8 Funding Statement

This study received no external funding and was conducted independently by the Author.

0.9 Conflict of Interest Statement

There is no conflicts of Interest related to this study.

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